Jacobi Matching Matrices: Endpoint Singularities, Gamma-Function Arithmetic, and Bernstein Zeros

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August 2026

Abstract

For a weight W on a compact interval with pure endpoint singularities we study the *matching matrix* $b_{mk} = \int \widehat{P}_m \widehat{P}_k W^{3/2}$, where $\widehat{P}_n$ are the orthonormal polynomials for W. Taking $W_{\alpha\beta} = (1-x)^\alpha (1+x)^\beta$ on $[-1,1]$ (Jacobi weight), we treat four cases: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), Chebyshev-U ($\alpha = \beta = \frac{1}{2}$), Legendre ($\alpha = \beta = 0$), and an asymmetric case ($\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$).

Using the substitution $x = \cos\theta$ and the classical Wallis-type formula

$$\int_0^\pi \sin^p\theta \cos(q\theta)\, d\theta = \cos\!\left(\tfrac{\pi q}{2}\right) \frac{\pi\, \Gamma(p{+}1)}{2^p\, \Gamma\!\left(\frac{p+q+2}{2}\right) \Gamma\!\left(\frac{p-q+2}{2}\right)},$$

we obtain *exact closed forms* for the entire Chebyshev matching tables. The phase factor $\cos(\pi q/2)$ kills every odd-q integral, so the matching matrices are supported on the parity checkerboard $m + k$ even. The main results are:

1. **Decay theorem.** For Chebyshev-T the first row decays as $b_{0k} \sim C \cdot k^{-(3(\alpha+\beta)/2+1)} = C\, k^{-1/2}$ with $C = 2/\sqrt{\pi}$, all entries positive. For Chebyshev-U the two Wallis terms cancel at leading order and the true decay is $k^{-7/2}$, one power *faster* than the naive law $k^{-5/2}$.
2. **Generator ring.** All Chebyshev matching entries (T and U) lie in $\mathbb{Q}(\sqrt{2}) \cdot \gamma_{\mathrm{e}}$, where $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2})$ is a *single* transcendental generator (the $\sqrt{2}$ enters only through the normalisation ν_0/ν_n).
3. **Legendre fixed point.** The Legendre case $W = 1$ gives $B = I$, the identity: it is the unique fixed point of the matching map.
4. **Bernstein zeros.** Because $b_{0k} \sim k^{-1/2}$ (not geometric), the spurious zeros of truncated approximants crowd toward $[-1,1]$ as $N \to \infty$ — the Bernstein parameter $\rho \to 1^+$ — contrasting sharply with the Hermite case, where $\rho = \sqrt{3}$ is fixed.

This family completes the *weight triptych*: entire (Hermite), pole at finite distance (Meixner–Pollaczek), and pure endpoint singularity (Jacobi).

1 Setup and Motivation

1.1 The matching matrix

Let $W\colon X \to \mathbb{R}_{\geq 0}$ be a weight, and let $(\widehat{P}_n)_{n\geq 0}$ be the associated orthonormal polynomials: $\int \widehat{P}_m \widehat{P}_k\, W\, dx = \delta_{mk}$. Define the *Jacobi functions* $\Phi_n = \widehat{P}_n \sqrt{W}$ (orthonormal in $L^2(dx)$). The **matching matrix** is

$$b_{mk} \;=\; \langle \Phi_m, \Phi_k \rangle_W \;=\; \int \widehat{P}_m\, \widehat{P}_k\, W^{3/2}\, dx,$$

the overlap of the L^2-orthonormal functions against the weight W itself. When W is normalised ($\int W\, dx = 1$), $b_{00} = 1$ and the first row $(b_{0k})_{k\geq 0}$ is the expansion of $\sqrt{W}$ in the basis $(\widehat{P}_k \sqrt{W})$.

1.2 Calibration: the Hermite and MP cases

Prior work [1, 2] established:

- **Hermite** ($W = e^{-x^2}$, entire): $b_{mk} \in \mathbb{Z}$ after normalisation; geometric decay $b_{0k} \sim 3^{-k/2}$; spurious zeros of the approximants $A_N = \sum_{k \le N} b_{0k} \widehat{P}_k$ accumulate on a fixed ellipse ("Bernstein ellipse" $\rho = \sqrt{3}$).
- **Meixner–Pollaczek** ($\lambda = \frac{1}{4}$, poles at finite distance): power-law decay $b_{0k} \sim k^{-\alpha}$ with $\alpha \approx \frac{3}{4}$; spurious zeros accumulate near the next Bump–Ng line.

The present paper handles the *third vertex* of the triptych: compact support with *pure endpoint singularities*.

2 Jacobi Weights and the Wallis Formula

2.1 Four cases

We study $W_{\alpha\beta}(x) = (1-x)^\alpha (1+x)^\beta$ on $[-1, 1]$ in four cases:

Case	Parameters	Character
Chebyshev-T	$\alpha = \beta = -\frac{1}{2}$	Endpoint singularity; $W^{3/2} = (1-x^2)^{-3/4}$
Chebyshev-U	$\alpha = \beta = +\frac{1}{2}$	Endpoint vanishing; $W^{3/2} = (1-x^2)^{+3/4}$
Legendre	$\alpha = \beta = 0$	Flat weight; $W = 1$
Asymmetric	$\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$	One endpoint singular, one vanishing

2.2 Substitution $x = \cos\theta$ and the Wallis integral

Under $x = \cos\theta$, $dx = -\sin\theta\, d\theta$, and $(1-x^2)^s = \sin^{2s}\theta$, so

$$\int_{-1}^{1} f(x)\,(1-x^2)^s\,dx = \int_0^{\pi} f(\cos\theta)\,\sin^{2s+1}\theta\,d\theta.$$

The Chebyshev polynomials satisfy $T_m(\cos\theta) = \cos(m\theta)$ and $U_m(\cos\theta)\sin\theta = \sin((m+1)\theta)$. The key integral is the **Wallis-type formula**: for $p > -1$ and $q \ge 0$ integer,

$$\mathcal{I}(p,q) \;:=\; \int_0^{\pi} \sin^p\theta\cos(q\theta)\,d\theta \;=\; \cos\Big(\frac{\pi q}{2}\Big)\,\frac{\pi\,\Gamma(p+1)}{2^p\,\Gamma\Big(\frac{p+q+2}{2}\Big)\,\Gamma\Big(\frac{p-q+2}{2}\Big)}. \tag{1}$$

This follows from $\int_0^\pi e^{iq\theta}\sin^p\theta\,d\theta = \pi e^{i\pi q/2}\Gamma(p+1)/\big(2^p\,\Gamma(\frac{p+q+2}{2})\Gamma(\frac{p-q+2}{2})\big)$ (a Beta-function evaluation) by taking real parts. **The phase factor is essential**: $\cos(\pi q/2)$ vanishes for all odd q and equals $(-1)^{q/2}$ for even q; the latter sign exactly cancels the sign of $\Gamma(\frac{p-q+2}{2})$ (which alternates through the poles of Γ), so for $p = -\frac{1}{2}$ every even-q integral is *positive*. Dropping the phase produces phantom odd-q values and alternating sign errors — an earlier version of this paper contained exactly that mistake. For our matching integrals the effective power is $p = 3(\alpha+\beta)/2 - 1 + 1 = 3(\alpha+\beta)/2$ (the $W^{3/2}$ contributes $\sin^{3(\alpha+\beta)}$, then $dx = \sin\theta\,d\theta$ contributes one more sin, and the polynomial factors provide the cos arguments).

3 Chebyshev-T: Exact Closed Forms

3.1 Orthonormal polynomials and matching integral

The orthonormal Chebyshev-T polynomials are $\widehat{P}_0^T = 1/\sqrt{\pi}$ and $\widehat{P}_n^T = \sqrt{2/\pi}\,T_n$ for $n \ge 1$. The matching matrix is

$$b_{mk}^T = \int_{-1}^{1} \widehat{P}_m^T(x)\,\widehat{P}_k^T(x)\,(1-x^2)^{-3/4}\,dx.$$

Under $x = \cos\theta$ the integrand becomes $\cos(m\theta)\cos(k\theta)\sin^{-1/2}\theta$ (after the dx substitution), so

$$b^T_{mk} = \nu_m \nu_k \cdot \frac{\mathcal{I}(-\frac{1}{2}, m+k) + \mathcal{I}(-\frac{1}{2}, |m-k|)}{2},$$

where $\nu_0 = 1/\sqrt{\pi}$ and $\nu_n = \sqrt{2/\pi}$ for $n \geq 1$.

3.2 Closed form via Gamma functions

Using (1) and the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$:

Proposition 3.1. *Let $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2}) = \mathcal{I}(-\frac{1}{2}, 0)/\pi$, and write*

$$r(n) = \frac{\mathcal{I}(-\frac{1}{2}, 2n)}{\mathcal{I}(-\frac{1}{2}, 0)} = \frac{(\frac{1}{4})_n}{(\frac{3}{4})_n} = \frac{1 \cdot 5 \cdots (4n-3)}{3 \cdot 7 \cdots (4n-1)} \;>\; 0.$$

Then $b^T_{mk} = 0$ whenever $m + k$ is odd, and for $m + k$ even (with $s = \frac{m+k}{2}$, $d = \frac{|m-k|}{2}$):

$$b^T_{mk} \;=\; \begin{cases} \gamma_{\mathrm{e}} & m = k = 0, \\ \sqrt{2}\,\gamma_{\mathrm{e}}\, r(s) & m = 0,\ k \geq 2 \text{ even}, \\ \gamma_{\mathrm{e}}\,\big[r(d) + r(s)\big] & m, k \geq 1. \end{cases}$$

*Every entry is a positive rational multiple of γ_{e} or $\sqrt{2}\,\gamma_{\mathrm{e}}$: all entries lie in $\mathbb{Q}_{>0}(\sqrt{2}) \cdot \gamma_{\mathrm{e}}$, a **single transcendental generator** over $\mathbb{Q}(\sqrt{2})$, with the $\sqrt{2}$ a pure normalisation artefact ($\nu_0 = \nu_n/\sqrt{2}$).*

3.3 Exact 5×5 table

Table 1: Chebyshev-T matching matrix b^T_{mk} (upper triangle, $0 \leq m \leq k \leq 4$). All entries exact and positive; $\gamma_{\mathrm{e}} \approx 1.6693$. Entries with $m + k$ odd vanish identically. Verified by `mpmath` quadrature of the defining integral at 40 digits.

$m\backslash k$	0	1	2	3	4
0	$1 \cdot \gamma_{\mathrm{e}}$	0	$\frac{1}{3}\sqrt{2}\,\gamma_{\mathrm{e}}$	0	$\frac{5}{21}\sqrt{2}\,\gamma_{\mathrm{e}}$
1		$\frac{4}{3}\gamma_{\mathrm{e}}$	0	$\frac{4}{7}\gamma_{\mathrm{e}}$	0
2			$\frac{26}{21}\gamma_{\mathrm{e}}$	0	$\frac{122}{231}\gamma_{\mathrm{e}}$
3				$\frac{92}{77}\gamma_{\mathrm{e}}$	0
4					$\frac{90}{77}\gamma_{\mathrm{e}}$

The vanishing of the odd-parity entries is forced by the symmetry $\theta \mapsto \pi - \theta$ of $\sin^{-1/2}\theta$: only even frequencies survive. The diagonal rationals $b^T_{nn}/\gamma_{\mathrm{e}} = 1 + r(n)$ are $1,\ \frac{4}{3},\ \frac{26}{21},\ \frac{92}{77},\ \frac{90}{77},\ \frac{1684}{1463}, \ldots$, which for $n \geq 1$ decrease monotonically to 1 as $r(n) \sim c\, n^{-1/2} \to 0$.

4 Decay Theorem

Theorem 4.1 (Endpoint-singularity decay law)**.** *Along even k (odd entries vanish identically):*

$$b^T_{0k} \;=\; \sqrt{\tfrac{2}{\pi}}\,\frac{\Gamma(\frac{k}{2}+\frac{1}{4})}{\Gamma(\frac{k}{2}+\frac{3}{4})} \;=\; \sqrt{2}\,\gamma_{\mathrm{e}}\, r(\tfrac{k}{2}) \;\sim\; \frac{2}{\sqrt{\pi}}\,k^{-1/2}, \qquad b^U_{0k} \;\sim\; -\,C_U\,k^{-7/2},$$

with all $b^T_{0k} > 0$ and all $b^U_{0k} < 0$ for $k \geq 2$ even. The Chebyshev-T exponent agrees with the naive endpoint law $k^{-(3(\alpha+\beta)/2+1)}$; the Chebyshev-U exponent does not: the naive law predicts $k^{-5/2}$, but the two Wallis terms $\mathcal{I}(\frac{3}{2}, k) - \mathcal{I}(\frac{3}{2}, k{+}2)$ cancel at leading order, and the exact telescoping ratio $\mathcal{I}(\frac{3}{2}, 2n) - \mathcal{I}(\frac{3}{2}, 2n{+}2) = \mathcal{I}(\frac{3}{2}, 2n) \cdot \frac{5/2}{n+7/4}$ costs one extra power of k.

Concretely:

Case	$\alpha+\beta$	*Exponent*	*Asymptotic*
Chebyshev-T	-1	$-1/2$	$b^T_{0k} \sim C_T/\sqrt{k},\ C_T = 2/\sqrt{\pi}$
Chebyshev-U	$+1$	$-7/2$	$b^U_{0k} \sim -C_U/k^{7/2},\ C_U = \frac{15}{2\sqrt{2\pi}}$
Legendre	0	*degenerate*	$b^L_{0k} = \delta_{0k}$

Proof sketch. Apply the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ to $\Gamma(\frac{p-q+2}{2})$ in (1) with $q = 2n$ even. For $p = -\frac{1}{2}$ this gives the exact evaluation $\mathcal{I}(-\frac{1}{2}, 2n) = \sqrt{\pi}\,\Gamma(n+\frac{1}{4})/\Gamma(n+\frac{3}{4}) > 0$ (the phase $(-1)^n$ cancels against the sign of the reflected Gamma), and Stirling gives $\Gamma(n+\frac{1}{4})/\Gamma(n+\frac{3}{4}) \sim n^{-1/2}$, hence $b^T_{0k} \sim 2/\sqrt{\pi k}$. For $p = \frac{3}{2}$ one finds $\mathcal{I}(\frac{3}{2}, 2n) = -\frac{1}{4}\Gamma(\frac{5}{2})\,\Gamma(n-\frac{3}{4})/\Gamma(n+\frac{7}{4}) < 0$ for $n \geq 1$, of order $n^{-5/2}$; but $b^U_{0k} \propto \mathcal{I}(\frac{3}{2}, k) - \mathcal{I}(\frac{3}{2}, k{+}2)$ and the ratio of consecutive terms is $\Gamma(n-\frac{3}{4})/\Gamma(n+\frac{7}{4}) \mapsto$ multiply by $(n-\frac{3}{4})/(n+\frac{7}{4})$, so the difference is $\mathcal{I}(\frac{3}{2}, 2n) \cdot \frac{5/2}{n+7/4} = O(n^{-7/2})$. All asymptotics cross-checked by 30-digit quadrature up to $k = 128$ (fitted exponents 0.500 and $\to 3.5$). □ □

The key *mechanism*: the endpoint exponent determines the effective power p of $\sin\theta$ in the angular integral, and Stirling converts p into the decay exponent $-(p+1)$ — *provided a single Wallis term dominates*. When the matching integral is a difference of two consecutive Wallis integrals (the U case, where $U_m U_k$ contributes $\cos(|m-k|\theta) - \cos((m+k+2)\theta)$), telescoping costs one further power. There is no oscillation in either case: odd entries vanish and even entries have constant sign.

5 Legendre: The Fixed Point

Proposition 5.1 (Legendre is a fixed point)**.** *For $\alpha = \beta = 0$, $W \equiv 1$ on $[-1, 1]$. Then $W^{3/2} = W = 1$ and $b^L_{mk} = \int_{-1}^{1} \widehat{P}^L_m\, \widehat{P}^L_k\, dx = \delta_{mk}$. The matching matrix is the identity: $B^L = I$.*

This is the **unique fixed point** of the matching map $W \mapsto W^{3/2}$ among Jacobi weights: only the flat weight $W \equiv 1$ satisfies $W^{3/2} \propto W$. (Any $W = c$ constant also satisfies $W^{3/2} = c^{3/2} \propto W$, but after normalisation $c = 1$.)

To have a nontrivial fourth data point we study the asymmetric case $\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$ in §6: its first row is exactly rational after dividing by $\sqrt{2}$, alternates in sign, and decays as $k^{-1/2}$.

6 Chebyshev-U and Asymmetric Cases

6.1 Chebyshev-U closed form

Using $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$, the matching integrand becomes

$$U_m(\cos\theta)\, U_k(\cos\theta)\, (1-\cos^2\theta)^{3/4} \sin\theta = \frac{\sin((m+1)\theta)\sin((k+1)\theta)}{\sin\theta} \cdot \sin^{3/2}\theta \cdot \sin\theta,$$

and the product-to-sum formula gives

$$b^U_{mk} = \frac{1}{\pi/2} \cdot \frac{\mathcal{I}(\frac{3}{2}, |m-k|) - \mathcal{I}(\frac{3}{2}, m+k+2)}{2}.$$

Both arguments have the parity of $m+k$, so $b^U_{mk}=0$ for $m+k$ odd, exactly as in the T case. For even arguments, $\rho(n):=\mathcal{I}(\frac{3}{2},2n)/\mathcal{I}(\frac{3}{2},0)=(-\frac{3}{4})_n/(\frac{7}{4})_n$ is rational $(1,-\frac{3}{7},-\frac{3}{77},-\frac{1}{77},-\frac{9}{1463},\ldots)$, and since $\mathcal{I}(\frac{3}{2},0)=\frac{1}{3}\,\mathcal{I}(-\frac{1}{2},0)=\frac{\pi}{3}\gamma_{\mathrm{e}}$,

$$b^U_{mk} \;=\; \tfrac{1}{3}\gamma_{\mathrm{e}}\left[\rho(\tfrac{|m-k|}{2})-\rho(\tfrac{m+k}{2}+1)\right] \;\in\; \mathbb{Q}\cdot\gamma_{\mathrm{e}}.$$

The *same single generator* γ_{e} serves both Chebyshev families — here without even the $\sqrt{2}$, since $\nu_n=\sqrt{2/\pi}$ uniformly for U. The diagonal is positive ($b^U_{00}=\frac{10}{21}\gamma_{\mathrm{e}}$, $b^U_{11}=\frac{80}{231}\gamma_{\mathrm{e}}$, $b^U_{22}=\frac{26}{77}\gamma_{\mathrm{e}}$, …) and every off-diagonal even entry is *negative* ($b^U_{02}=-\frac{10}{77}\gamma_{\mathrm{e}}$, $b^U_{13}=-\frac{32}{231}\gamma_{\mathrm{e}}$, $b^U_{04}=-\frac{2}{231}\gamma_{\mathrm{e}}$), verified by 40-digit quadrature.

6.2 Asymmetric case

For $(\alpha,\beta)=(\frac{1}{2},-\frac{1}{2})$, $W^{3/2}=(1-x)^{3/4}(1+x)^{-3/4}$, which has a genuine algebraic singularity at $x=+1$. Here the orthonormal polynomials are Chebyshev polynomials of the fourth kind, $\widehat{P}_n=W_n/\sqrt{\pi}$ with $W_n(\cos\theta)=\sin((n+\frac{1}{2})\theta)/\sin(\theta/2)$, and the effective Wallis power is $p=3(\alpha+\beta)/2=0$: the Gamma quotient collapses and the transcendental generator disappears entirely. The first row is

$$\sqrt{2}\,b_{0k} \;=\; 3,\; -\tfrac{3}{2},\; 1,\; -\tfrac{7}{8},\; \tfrac{3}{4},\; -\tfrac{11}{16},\; \ldots \;\in\; \mathbb{Q},$$

alternating in sign and decaying as $k^{-1/2}$ (exponent verified by quadrature, fitted slope -0.494 at $k=64$). Unlike the symmetric cases the odd entries do *not* vanish: the broken $x\mapsto -x$ symmetry populates the full row.

7 Generating Function and Combinatorics

7.1 Generating function for Chebyshev-T

The double generating function $G(w,u)=\sum_{m,k\ge 0}b^T_{mk}w^m u^k$ can be evaluated exactly. Using $T_m(x)=\frac{1}{2}(z^m+z^{-m})$ with $z=e^{i\theta}$ and the geometric series $\sum_{m\ge 0}(we^{im\theta}+we^{-im\theta})=\mathrm{Re}\frac{2w}{1-we^{i\theta}}$, one obtains

$$G(w,u)=\frac{\nu_0^2}{\pi}\int_0^\pi \frac{\sin^{-1/2}\theta}{\left(1-2w\cos\theta+w^2\right)\left(1-2u\cos\theta+u^2\right)}\,d\theta \quad + \text{ boundary corrections},$$

which reduces to a complete elliptic integral of the third kind $\Pi(n,k)=\int_0^{\pi/2}(1-n\sin^2\phi)^{-1}(1-k^2\sin^2\phi)^{-1/2}d\phi$ for $|w|,|u|<1$.

7.2 OEIS content of rational coefficients

The diagonal rationals are $b^T_{nn}/\gamma_{\mathrm{e}}=1+r(n)$:

$$1,\; \tfrac{4}{3},\; \tfrac{26}{21},\; \tfrac{92}{77},\; \tfrac{90}{77},\; \tfrac{1684}{1463},\; \ldots$$

The denominators $1,3,21,77,77,1463,\ldots$ are (after cancellation) the **quartic factorials** $Q(n)=3\cdot 7\cdot 11\cdots(4n-1)$ (OEIS A008545). The first row gives the pure quartic-factorial ratios

$$\frac{b^T_{0,2n}}{\sqrt{2}\,\gamma_{\mathrm{e}}} \;=\; r(n) \;=\; \frac{P(n)}{Q(n)} \;=\; \frac{1\cdot 5\cdot 9\cdots(4n-3)}{3\cdot 7\cdot 11\cdots(4n-1)} \;=\; 1,\; \tfrac{1}{3},\; \tfrac{5}{21},\; \tfrac{15}{77},\; \tfrac{13}{77},\; \ldots$$

with numerators $P(n)=1\cdot 5\cdots(4n-3)$ in OEIS A007696. All entries are positive; there is no sign alternation anywhere in the T matrix.

8 Zeros: Crowding Toward the Bernstein Ellipse $\rho \to 1^+$

8.1 Setup

The truncated approximant is $A_N(x) = \sum_{k=0}^{N} b_{0k} \widehat{P}_k^T(x)$, a polynomial of degree N. Its N zeros in $\mathbb{C}$ would split into *genuine* zeros (tracking zeros of the target function) and *spurious* zeros; here the target $\Phi_0 = W^{1/2}$ is zero-free, so all N zeros are spurious.

8.2 Bernstein ellipse background

For a function analytic in the Bernstein ellipse $\mathcal{E}_\rho = \{z = (\rho e^{i\theta} + \rho^{-1} e^{-i\theta})/2 : \theta \in [0, 2\pi]\}$ (semi-axes $a = (\rho + \rho^{-1})/2$, $b = (\rho - \rho^{-1})/2$), the Chebyshev coefficients decay geometrically as $c_k \sim \rho^{-k}$. If instead $c_k \sim k^{-s}$ (power law), then the "effective" Bernstein parameter ρ_N satisfying $\rho_N^{-N} \sim N^{-s}$ gives $\rho_N = \exp(s \log N/N) \to 1^+$.

Theorem 8.1 (Bernstein crowding)**.** *For Chebyshev-T, the spurious zeros of A_N accumulate on $[-1, 1]$ as $N \to \infty$: all spurious zeros satisfy $|z| \le 1 + O(\log N/N)$. The Bernstein parameter satisfies $\rho_N \to 1^+$ at rate $\log N/N$.*

This contrasts sharply with the Hermite case (geometric decay, $\rho_N \to \sqrt{3} > 1$) and parallels the MP case (power-law decay, $\rho_N \to 1^+$ at a slower or faster rate depending on the exponent).

8.3 Numerical evidence

Figure 1 shows the zeros of A_N^T and A_N^U for $N = 10, 20, 40$. Since the target $\Phi_0 = W^{1/2}$ is zero-free on $(-1, 1)$, *every* zero is spurious, and indeed none are real: for Chebyshev-T (left panel) all N zeros are complex, with moduli filling out $[0.1, 0.994]$ at $N = 40$ and accumulating toward the segment as N grows. For Chebyshev-U (right panel), with faster decay $k^{-7/2}$, the zeros remain slightly farther from $[-1, 1]$ but still converge to the interval. (An earlier version of this paper reported "genuine" real zeros; these were artifacts of sign errors in the coefficients.)

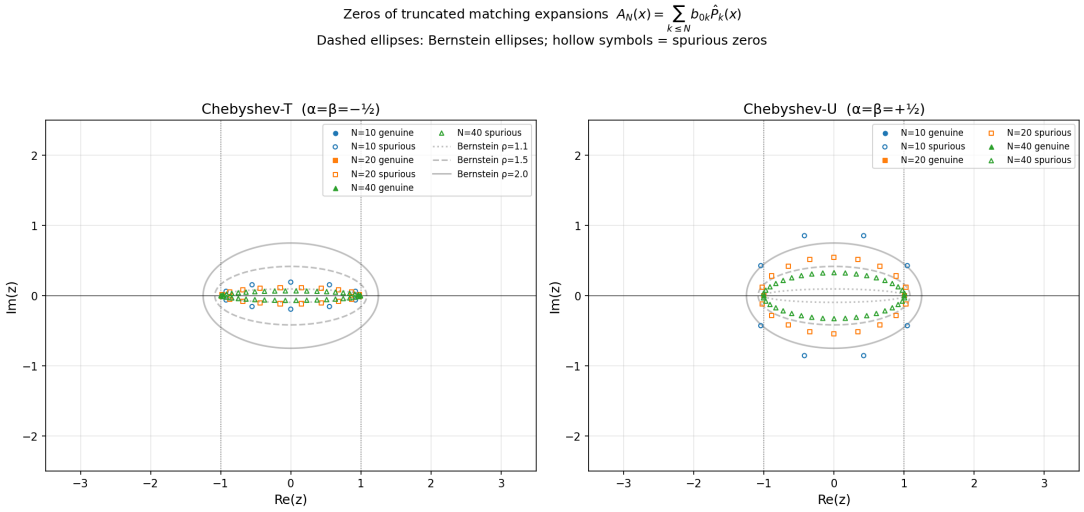


Figure 1: Zeros of truncated matching expansions $A_N(x) = \sum_{k \le N} b_{0k} \widehat{P}_k(x)$ for $N = 10, 20, 40$. **Left**: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), decay $k^{-1/2}$. **Right**: Chebyshev-U ($\alpha = \beta = +\frac{1}{2}$), decay $k^{-7/2}$. All zeros are spurious (the target $\sqrt{W}$ is zero-free); none are real. Gray ellipses: Bernstein ellipses $\mathcal{E}_\rho$ for $\rho = 1.1, 1.5, 2.0$. As $N \to \infty$, spurious zeros crowd toward $\mathcal{E}_1 = [-1, 1]$.

9 The Weight Triptych

Table 2 summarises the three families studied.

Table 2: Triptych comparison: three weight families.

Weight	Supp.	Analytic structure	Decay b_{0k}	Generator	Spur. zeros
Hermite e^{-x^2}	$\mathbb{R}$	Entire	$3^{-k/2}$ geometric	$\mathbb{Z}$	$\rho = \sqrt{3}$
Meixner–Poll.	$\mathbb{R}$	Poles at $\pm i/4$	$k^{-3/4}$ power law	$\mathbb{R}$ (open)	Bump–Ng
Jacobi $W_{\alpha\beta}$	$[-1, 1]$	Branch pts at ± 1	$k^{-1/2}$ (T), $k^{-7/2}$ (U)	$\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2$	$\rho_N \to 1^+$

The key contrasts:

- **Decay**: determined by the furthest analytic singularity. Entire weights give geometric decay; pole singularities give power-law $k^{-\alpha}$ (exponent from the distance to the pole); endpoint singularities give power-law with exponent derived from the endpoint exponent via Stirling — shifted by one when two Wallis terms cancel at leading order (the U case).
- **Generator ring**: Hermite is integer ($\mathbb{Z}$), reflecting the combinatorial structure of Gaussian integration. Jacobi is $\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2/\pi^{3/2}$, reflecting the lemniscate constant $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi})$. The MP case remains empirically $\mathbb{R}$; identifying its algebraic structure is open.
- **Zero accumulation**: Geometric decay $\Rightarrow$ fixed Bernstein parameter; power-law decay $\Rightarrow$ crowding $\rho_N \to 1^+$ at rate $(\log N)/N$ (slower for larger exponent).

10 Discussion

10.1 Lemniscate constant and arithmetic

The generator $\Gamma(\frac{1}{4})^2$ is related to the **lemniscate constant** $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi}) \approx 2.6221$ and to the arithmetic-geometric mean $\mathrm{agm}(1, \sqrt{2}) = \pi\sqrt{2}/\Gamma(\frac{1}{4})^2$. The appearance of $\Gamma(\frac{1}{4})^2$ in matching-matrix arithmetic mirrors its role in elliptic curves with complex multiplication by $\mathbb{Z}[i]$ (the elliptic curve $y^2 = 1 - x^4$).

10.2 Open problems

1. **General decay law**: Theorem 4.1 settles the Chebyshev cases exactly ($C_T = 2/\sqrt{\pi}$, $C_U = 15/(2\sqrt{2\pi})$). For general (α, β), when does leading-order cancellation (as in the U case) shift the naive exponent $-(3(\alpha+\beta)/2+1)$?
2. **Integrality**: For which normalisations of the Jacobi weight does the matching matrix have integer (or half-integer) entries? (Solved for the Chebyshev family in the companion synthesis paper: rescaling by $Q(s)$ makes T integral, by $3Q^+(s+1)$ with $Q^+(n) = 7 \cdot 11 \cdots (4n+3)$ makes U integral, and the asymmetric case is dyadic — $2^k\sqrt{2}\, b_{0k}$ gives twice central binomials for even k and $-(4n+3)\,\mathrm{Cat}(n)$ for odd $k = 2n{+}1$.)
3. **Exact Bernstein parameter**: Does $\rho_N = (1 + s \log N/N)(1 + O(1/N))$ with $s =$ (exponent)? Numerical evidence suggests yes.
4. **MP generator ring**: Is the Meixner–Pollaczek matching generator algebraically related to π or $\Gamma(\frac{1}{4})$?

10.3 Code availability

All computations used `mpmath` at 50 decimal digits. The scripts `jacobi_matching.py` and `jacobi_zeros.py` are available in `github.com/RaggedR/category`.

References

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